

VPN and Remote Access Security — Guided Notes

ANSWER KEY

Unit 4.2.2 | CompTIA Network+ N10-009 Obj. 4.3 | N10-008 4.3

For instructor use / distribute after students complete the notes.

Question 1

IPsec uses two protocols: _____ (AH) provides authentication and integrity but no encryption, and _____ (ESP) provides encryption, authentication, and integrity. Which is used in modern VPN deployments and why?

Answer: Authentication Header; Encapsulating Security Payload. ESP is used because it provides confidentiality — without encryption, AH alone exposes the payload to eavesdropping.

Question 2

IPsec tunnel mode encapsulates the _____ original IP packet inside a new IP packet. Transport mode protects only the _____, leaving the original IP header visible. Which mode is standard for site-to-site VPNs?

Answer: Entire; payload. Tunnel mode — the VPN gateways add new outer headers to carry traffic across the internet while hiding the original source/destination.

Question 3

IKE Phase 1 establishes a _____ channel used for negotiation, authenticating peers via _____ or certificates. IKE Phase 2 uses this channel to negotiate the _____ that will protect actual data traffic.

Answer: Secure management (ISAKMP SA); pre-shared keys (PSK); IPsec Security Associations (SAs).

Question 4

SSL/TLS VPNs operate over TCP port _____, which makes them traverse most firewalls and NAT devices easily. IPsec VPNs can struggle with NAT because NAT modifies _____, breaking IPsec's integrity verification.

Answer: 443; IP headers.

Question 5

In a full-tunnel VPN, _____ client traffic flows through the VPN. In a split-tunnel VPN, only traffic destined for _____ routes through the tunnel. What is the security tradeoff of split tunnel?

Answer: All; internal (organizational) networks. Split tunnel reduces bandwidth but leaves internet-bound traffic unmonitored by the organization's security controls.

Question 6

Multi-factor authentication (MFA) for VPN access requires something you _____ plus something you _____. Why is username and password alone insufficient for VPN access?

Answer: Know (password); have (token, phone, hardware key). Credentials are frequently stolen via phishing or breaches — a second factor means a stolen password alone cannot grant VPN access.

Question 7

Endpoint posture checks evaluate the _____ device before granting full VPN access. List two conditions a posture check might require before allowing connection.

Answer: Client (connecting); examples: OS is fully patched, antivirus is installed and current, local firewall is enabled, device is domain-joined or MDM-enrolled.

Question 8 — Real World Application

REAL WORLD: A former contractor's VPN account was never deactivated. Three months after their contract ended, unauthorized access to internal project files is discovered through that account. Identify two process failures that allowed this and state the control that prevents each.

Answer: Failure 1: No offboarding procedure to revoke VPN access — fix: documented offboarding checklist that includes immediate VPN account deactivation. Failure 2: No periodic access review to catch stale accounts — fix: quarterly account audits that remove inactive credentials.